

CHRISTOPHER B. ENGBORG

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EDUCATION

University of Connecticut

Storrs, CT

Master's student in Physiology and Neurobiology

2022-20224

Advisor: Natale R. Sciolino, PhD

Anatomy and function of the cannabinoid receptor type 1 within the locus coeruleus of mice

BS in Molecular & Cell Biology

2018-2022

RESEARCH EXPERIENCE

University of Connecticut

Storrs, CT

PhD student, Physiology and Neurobiology

12/2024-present

Exploring the effects of multiple timescales of synaptic plasticity in models of spiking recurrent neural networks.

University of Connecticut

Storrs, CT

Master's thesis, Physiology and Neurobiology

08/2022-12/2024

Characterizing the anatomical and behavioral effects of loss of cannabinoid receptor type 1 (CB1) in noradrenergic neurons using a mouse genetics strategy.

Undergraduate Researcher

08/2021-08/2022

Designed and analyzed a fiber photometry experiment to determine the impact of sensory stimuli on the locus coeruleus. Additionally, helped establish the lab by assembling and automating optogenetic, fiber photometry, and operant conditioning systems.

TECHNICAL SKILLS

In situ hybridization, immunohistochemistry, cryosectioning, epifluorescent and confocal microscopy, mouse behavior, in-vivo fiber photometry, stereotaxic surgery for viral injection and optical implantation, computer programming (Python, MATLAB, R), mouse colony management, and PCR.

PUBLICATIONS

[Google Scholar](#), [ORCID](#), [NCBI](#)

1. Engborg, CB, Pujar, M, Kanadia, RN, & Sciolino, NR. (2025). Stress reactivity is modulated by cannabinoid type-1 receptors in norepinephrine and epinephrine neurons in

a context-dependent manner. *Neuroscience*, 585, 14–27. PubMed PMID: 40885375 (JCR impact factor 2.8).

2. Fan W, **Engborg, CB**, and Sciolino NR. Locus coeruleus dynamics are suppressed during licking and enhanced post-licking independent of taste novelty. (2024). *eNeuro*, 11 (4). PubMed PMID: 38649278 (JCR impact factor 4.4).

POSTERS

1. Bressler F, Fan W, **Engborg C**, Sciolino NR. Stimulation of the locus coeruleus to insular pathways reduced neuronal activity in the gustatory cortex. UConn Research Experience for Undergraduates Symposium, 2023.

GRANTS

1. **Institute for the Brain & Cognitive Sciences Seed Grant** Spring 2024
Title: Effects of in vivo endocannabinoid signaling in the locus coeruleus region during stress
Total: \$25,000
Role: Graduate project lead
Collaborator: Natale Sciolino (PI)
2. **Institute for the Brain & Cognitive Sciences Undergraduate Research Grant** Summer 2022
Title: *In vivo* fiber photometry of calcium dynamics in the locus coeruleus during sensory stimuli.
Total: \$5,000
Role: PI
Collaborator: Natale Sciolino (Faculty Supervisor)

TEACHING EXPERIENCE

University of Connecticut

Storrs, CT

1. **Enhanced Anatomy and Physiology Discussion** (PNB 2274) Fall 2023
Graduate Teaching Assistant, Instructors: Geoffrey Tanner and Xinnian Chen
(12 discussion session, office hours for ~30 students)
2. **Enhanced Anatomy and Physiology Lab** (PNB 2275) Spring 2023
Graduate Teaching Assistant. Instructors: Geoffrey Tanner and Xinnian Chen
(12 lab session, 12 quizzes, 1 essay, office hours for ~30 students)
3. **Enhanced Anatomy and Physiology Lab** (PNB 2274) Fall 2022
Graduate Teaching Assistant, Instructors: Geoffrey Tanner and Xinnian Chen
(12 lab session, 12 quizzes, 1 essay, office hours for ~30 students)
4. **Introduction to Biochemistry** (MCB 2000) Spring 2021 – Spring 2022
Supplemental Instructor, Instructor: Andrea Makkay
(28 independently designed lectures, office hours for ~40 students)

STUDENTS MENTORED

1. **Manaswini Pujar**, Undergraduate in Physiology and Neurobiology Major, Minor in Computer Science, Spring 2023-pres. 2024 Summer Undergraduate Research Fund (SURF) recipient.

SERVICE

1. *Ad hoc* reviewer for: *The Journal of NeuroPsychoPharmacology* (Co-submitted with Natale Sciolino)

PROFESSIONAL REFERENCES

1) **Scott Rich, PhD**

Assistant Professor, Department of Physiology & Neurobiology
University of Connecticut
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2) **Natale Sciolino, PhD**

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3) **Daniel Mulkey, PhD**

Professor & Associate Department Head, Department of Physiology & Neurobiology
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